pyDMS

Reproducible Dual-Mode Sorption Parameters

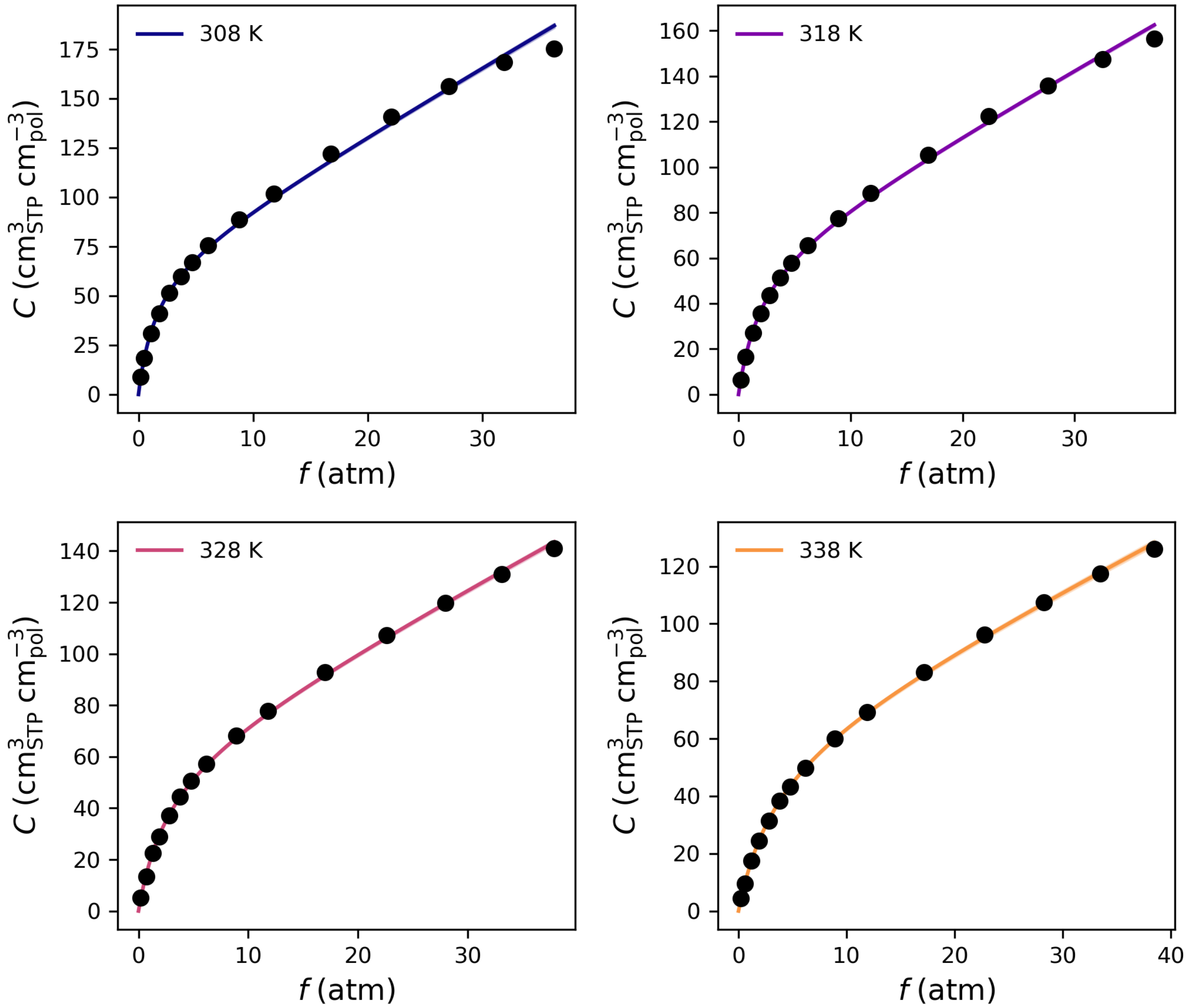
Analysis performed at 2025-06-13 18:09:27 Eastern Daylight Time

DMS Parameters

Temperature	Parameter	Result ± Error
308 K	k_D	3.368154088507561 ± 0.04993446030915689
	C'_H	67.40672715222968 ± 0.5719621482489714
	b	0.6481734250747876 ± 0.005900435837032988
318 K	k_D	2.749664738835181 ± 0.023904264708484384
	C'_H	63.883276956702886 ± 0.6222754809694209
	b	0.4816375021030929 ± 0.004189614484166503
328 K	k_D	2.272690165670627 ± 0.03240644964505831
	C'_H	61.44977555736838 ± 0.6980437212892951
	b	0.3644294372917349 ± 0.00345607239506266
338 K	k_D	1.8997504238198326 ± 0.04480433102322452
	C'_H	60.05939954323809 ± 0.8124343895662562
	b	0.2803320100569059 ± 0.0030958643645768104

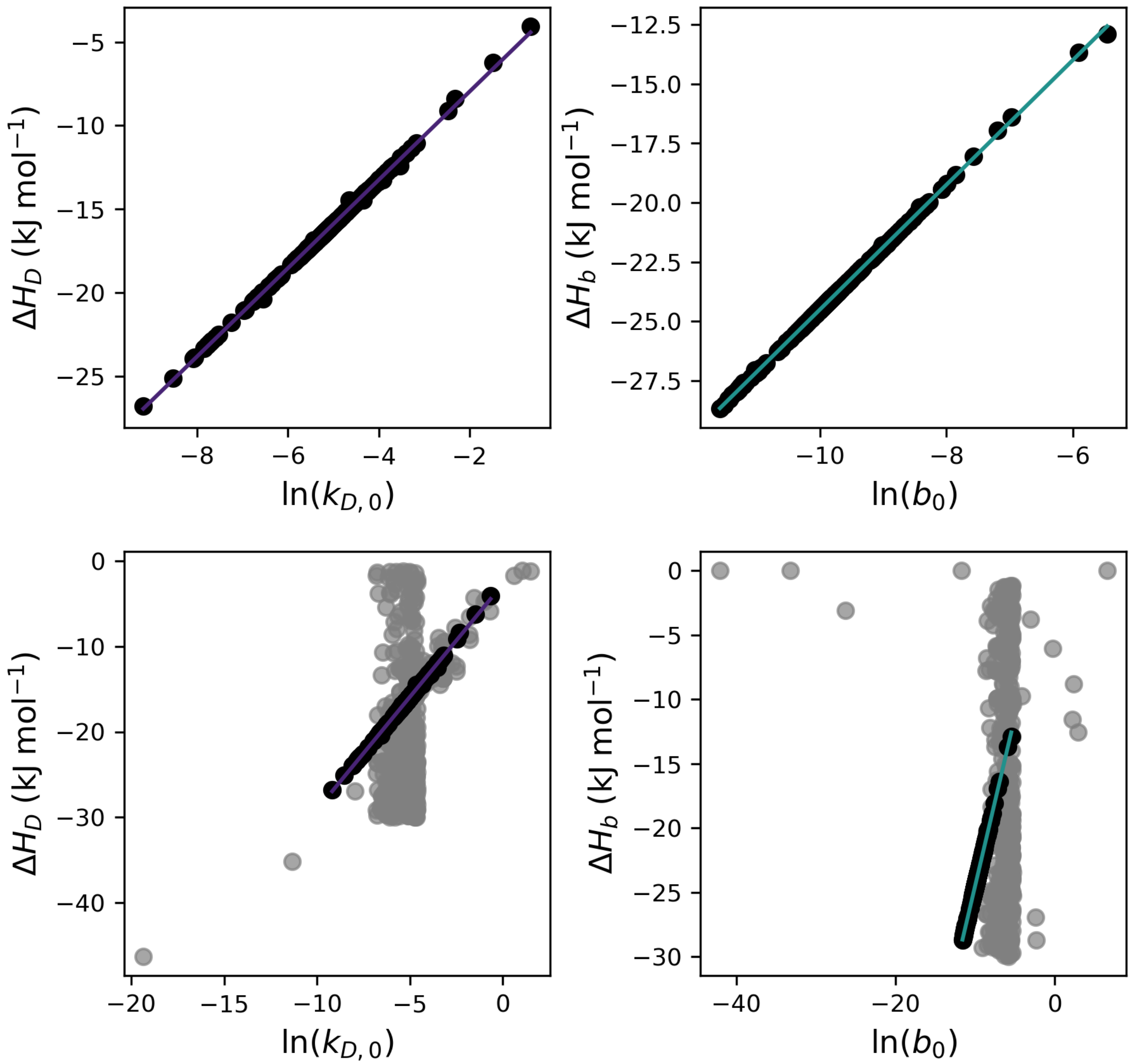
Zachary Smith RESEARCH LAB

Sorption Isotherms

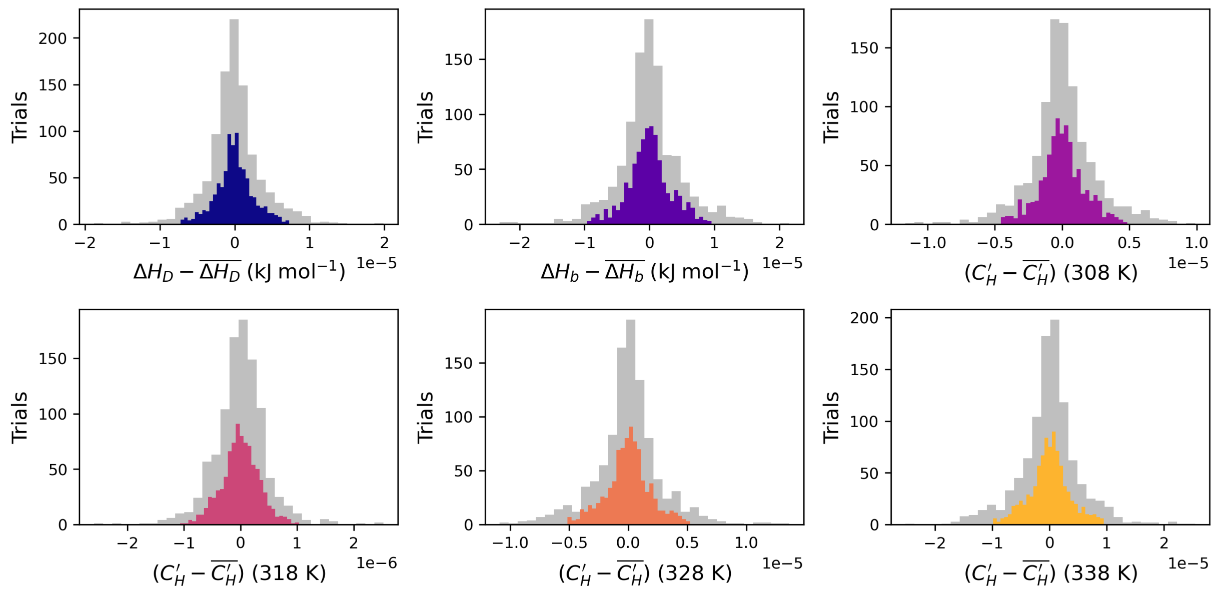


Linear Free Energy Relationships

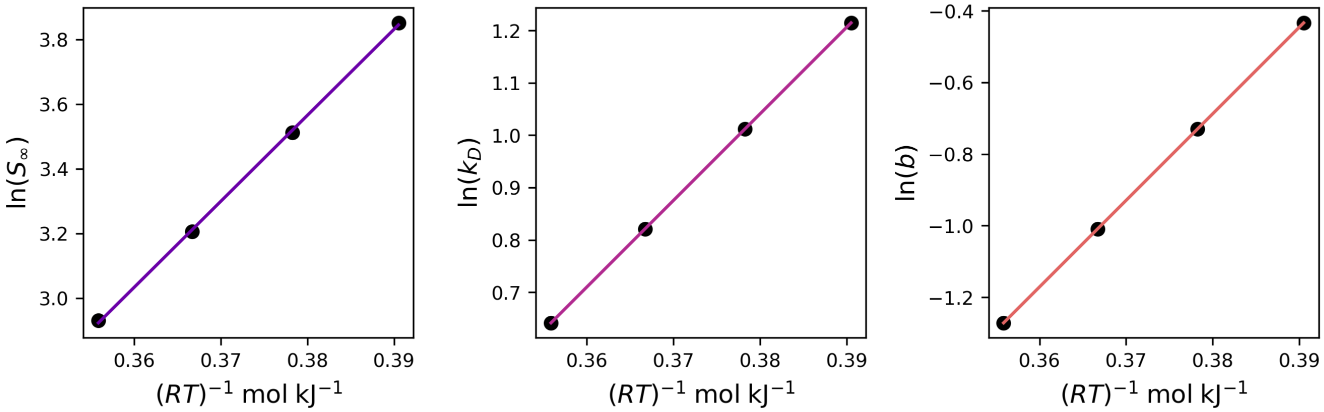
Heat of Henry Sorption (kJ/mol)	$\Delta H_D = -2.6747 + 2.6438 \ln(k_{D,0})$
Heat of Langmuir Sorption (kJ/mol)	$\Delta H_b = 1.7724 + 2.6277 \ln(b_0)$



van't Hoff Relationships



Sorption Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	1.449e-03±1.944e-04	-26.586097 ± 0.359713
i = Langmuir	5.314e-03 ± 6.053e-17	-16.521125 ± 0.978979
i = Henry	5.134e-05 ± 9.314e-19	-24.182175 ± 0.276747

Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-1, -30]
ΔH_b Initial Guess Range	[-1, -30]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-50, 0]
ΔH_b Solver Bounds	[-50, 0]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H #1	[0 150]
C'_H #2	[0 150]
C'_H #3	[0 150]
C'_H #4	[0 150]
Trials	1000
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-12
xtol (trust-constr)	1e-12
gtol (trust-constr)	1e-12

Questions or Comments?

Please reach out through the pyDMS GitHub page (*INSERT LINK*)

Citation

If you used pyDMS in any presented or published work please cite:
INSERT PAPER

License